

A.1 Proof of Lemma 3.1

Proof. Recall that $m_0 \in C^r(\mathcal{Z})$, where $\mathcal{Z} \subset \mathbb{R}^q$ is compact and $r \geq 2$. Let $\mathcal{M}_{K_n}$ denote the spline sieve space spanned by B-spline basis functions $\{B_1, \dots, B_{K_n}\}$ of order at least r , constructed over a quasi-uniform partition of $\mathcal{Z}$.

For notational clarity, consider first the case $q = 1$, i.e., $\mathcal{Z} = [0, 1]$. The multivariate case follows by tensor-product spline constructions with analogous rates. Let the interval $[0, 1]$ be partitioned into K_n subintervals with knots $0 = t_0 < t_1 < \dots < t_{K_n} = 1$, where the mesh size satisfies

$$h_n := \max_{1 \leq j \leq K_n} (t_j - t_{j-1}) \asymp K_n^{-1},$$

where $\asymp$ stands for *asymptotic equivalence*. Quasi-uniformity ensures that $h_n \leq CK_n^{-1}$ for some constant $C > 0$.

Since $m_0 \in C^r([0, 1])$, for each subinterval $[t_{j-1}, t_j]$ we may approximate m_0 by its Taylor polynomial of degree $r - 1$ around a point in the interval. By standard Taylor expansion with remainder, for any $z \in [t_{j-1}, t_j]$,

$$m_0(z) = P_{j,r-1}(z) + R_j(z),$$

where $P_{j,r-1}$ is a polynomial of degree $r - 1$ and the remainder satisfies $|R_j(z)| \leq Ch_n^r$, with C depending only on the supremum norm of the r -th derivative of m_0 . Hence, there exists a piecewise polynomial function p_n of degree $r - 1$ such that

$$\|m_0 - p_n\|_\infty \leq Ch_n^r.$$

The spline space $\mathcal{M}_{K_n}$ contains all piecewise polynomials of degree $r - 1$ with continuity constraints up to order $r - 2$. Classical results in spline approximation theory (see, e.g., de Boor (2001)[5]) guarantee that for any sufficiently smooth function, there exists a spline $m_{K_n} \in \mathcal{M}_{K_n}$ such that

$$\|p_n - m_{K_n}\|_\infty \leq Ch_n^r.$$

Combining with the previous step yields

$$\|m_0 - m_{K_n}\|_\infty \leq \|m_0 - p_n\|_\infty + \|p_n - m_{K_n}\|_\infty \leq Ch_n^r.$$

Since $h_n \asymp K_n^{-1}$, we obtain

$$\|m_0 - m_{K_n}\|_\infty \leq CK_n^{-r}.$$

For $q > 1$, the result follows by considering tensor-product B-spline bases over a quasi-uniform grid on $\mathcal{Z} \subset \mathbb{R}^q$. Under the assumption $m_0 \in C^r(\mathcal{Z})$, standard multivariate spline approximation results yield the same rate (up to constants depending on q), namely

$$\inf_{m \in \mathcal{M}_{K_n}} \|m_0 - m\|_\infty \leq CK_n^{-r}.$$

Thus, there exists a spline function $m_{K_n} \in \mathcal{M}_{K_n}$ satisfying

$$\|m_0 - m_{K_n}\|_\infty = O(K_n^{-r}),$$

which completes the proof. □

A.2 Proof of Lemma 3.2

Proof. Recall that the weights are defined as

$$w_i(\beta, m) = \frac{1}{C(X_i, Z_i; \beta, m)},$$

where $C(x, z; \beta, m) = \int_{a_Y}^{b_Y} f_\varepsilon(y - m(z) - g(x; \beta)) dy$.

By Assumption 4 (Truncation Stability), there exists a constant $c > 0$ such that

$$\inf_{x, z} C(x, z; \beta, m) \geq c$$

for all (β, m) in a neighborhood of (β_0, m_0) . In particular, this implies that for each observation (X_i, Z_i) ,

$$C(X_i, Z_i; \beta, m) \geq c \quad \text{almost surely.}$$

Using the definition of $w_i(\beta, m)$, we obtain

$$w_i(\beta, m) = \frac{1}{C(X_i, Z_i; \beta, m)} \leq \frac{1}{c} \quad \text{for all } i = 1, \dots, n.$$

Thus, it can be concluded that $\sup_{1 \leq i \leq n} w_i(\beta, m) \leq \frac{1}{c}$.

Since the upper bound $1/c$ is a deterministic constant independent of n , it follows immediately that

$$\sup_{1 \leq i \leq n} w_i(\beta, m) = O_p(1).$$

Because $(\hat{\beta}, \hat{m}) \rightarrow_p (\beta_0, m_0)$ and the truncation stability condition holds in a neighborhood of (β_0, m_0) , the same bound applies when (β, m) is replaced by $(\hat{\beta}, \hat{m})$ with probability tending to one. Hence the weights remain uniformly bounded in probability along the estimator sequence.

Therefore, we obtain that $\sup_{1 \leq i \leq n} w_i(\beta, m) = O_p(1)$, which completes the proof. $\square$

A.3 Proof of Lemma 3.3

Proof. Recall that the log-likelihood is given by

$$\ell_n(\beta, \theta) = \sum_{i=1}^n \log f_\varepsilon(Y_i - m(Z_i) - g(X_i; \beta)) - \sum_{i=1}^n \log C(X_i, Z_i; \beta, m),$$

where $m(z) = \sum_{k=1}^{K_n} \theta_k B_k(z)$. Let $S_{\theta, n} = \nabla_\theta \ell_n(\beta, \theta)$ denote the score vector with respect to θ .

Differentiating $\ell_n(\beta, \theta)$ with respect to θ_k , we obtain

$$\frac{\partial \ell_n}{\partial \theta_k} = \sum_{i=1}^n \left[-\dot{\ell}_\varepsilon(R_i(\beta, \theta)) B_k(Z_i) + \frac{\partial C(X_i, Z_i; \beta, m) / \partial \theta_k}{C(X_i, Z_i; \beta, m)} \right],$$

where $R_i(\beta, \theta) = Y_i - m(Z_i) - g(X_i; \beta)$ and $\dot{\ell}_\varepsilon(u) = \frac{d}{du} \log f_\varepsilon(u)$. Thus, one can express that $S_{\theta,n} = \sum_{i=1}^n \psi_i(\beta, \theta)$, where each $\psi_i(\beta, \theta)$ is a K_n -dimensional vector with k -th component

$$\psi_{i,k} = -\dot{\ell}_\varepsilon(R_i(\beta, \theta)) B_k(Z_i) + \frac{\partial C(X_i, Z_i; \beta, m) / \partial \theta_k}{C(X_i, Z_i; \beta, m)}.$$

By Assumption 5, $\dot{\ell}_\varepsilon(\cdot)$ is bounded. Since the spline basis functions are uniformly bounded on the compact domain $\mathcal{Z}$, there exists a constant C_1 such that $|\dot{\ell}_\varepsilon(R_i) B_k(Z_i)| \leq C_1$. Next, consider the second term. By differentiation under the integral sign,

$$\frac{\partial C(x, z; \beta, m)}{\partial \theta_k} = - \int_{a_Y}^{b_Y} f'_\varepsilon(y - m(z) - g(x; \beta)) B_k(z) dy.$$

Using boundedness of f'_ε and B_k , we obtain $\left| \frac{\partial C(x, z; \beta, m)}{\partial \theta_k} \right| \leq C_2$.

Under the truncation stability assumption, $C(x, z; \beta, m) \geq c > 0$, hence

$$\left| \frac{\partial C(x, z; \beta, m) / \partial \theta_k}{C(x, z; \beta, m)} \right| \leq \frac{C_2}{c}.$$

Therefore, each component $\psi_{i,k}$ is uniformly bounded, i.e., $|\psi_{i,k}| \leq C_3$. Because the observations are independent, we have $\text{Var}(S_{\theta,n}) = \sum_{i=1}^n \text{Var}(\psi_i)$. Each ψ_i is a K_n -dimensional vector with bounded components, so its covariance matrix has operator norm bounded by CK_n for some constant C . Hence,

$$\text{Var}(S_{\theta,n}) = O(nK_n).$$

Using the identity $\mathbb{E}\|S_{\theta,n}\|^2 = \text{trace}(\text{Var}(S_{\theta,n})) + \|\mathbb{E}S_{\theta,n}\|^2$, and noting that $\|\mathbb{E}S_{\theta,n}\| = O(n)$ but is dominated by variance after centering arguments (or working locally around the true parameter), we obtain

$$\mathbb{E}\|S_{\theta,n}\|^2 = O(nK_n).$$

Applying Markov's inequality (or Chebyshev's inequality) yields $\|S_{\theta,n}\| = O_p(\sqrt{nK_n})$.

The factor $\sqrt{n}$ reflects aggregation over independent observations, while the factor $\sqrt{K_n}$ reflects the dimensionality of the spline coefficient vector. Thus the score behaves like a K_n -dimensional random vector with variance proportional to n per coordinate. This completes the proof. $\square$

A.4 Proof of Theorem 3.4

Proof. Let

$$\ell_n(\beta, m) = \sum_{i=1}^n \log f_\varepsilon \{Y_i - m(Z_i) - g(X_i; \beta)\} - \sum_{i=1}^n \log C(X_i, Z_i; \beta, m)$$

denote the truncated log-likelihood, and define the normalized criterion

$$M_n(\beta, m) = \frac{1}{n} \ell_n(\beta, m).$$

The corresponding population criterion is

$$M(\beta, m) = E [\log f_\varepsilon \{Y - m(Z) - g(X; \beta)\} - \log C(X, Z; \beta, m)].$$

The proof proceeds by establishing uniform convergence of M_n to M , controlling the spline approximation error, and then invoking identifiability of the population maximizer.

Step 1: Decomposition of the empirical criterion.

Write

$$M_n(\beta, m) - M(\beta, m) = (P_n - P)\psi_{\beta, m} - (P_n - P)\rho_{\beta, m},$$

where

$$\psi_{\beta, m}(Y, X, Z) = \log f_\varepsilon \{Y - m(Z) - g(X; \beta)\}$$

and

$$\rho_{\beta, m}(X, Z) = \log C(X, Z; \beta, m).$$

Consequently,

$$\sup_{(\beta, m)} |M_n(\beta, m) - M(\beta, m)| \leq \sup_{(\beta, m)} |(P_n - P)\psi_{\beta, m}| + \sup_{(\beta, m)} |(P_n - P)\rho_{\beta, m}|.$$

The first term corresponds to the ordinary regression likelihood, whereas the second term is generated specifically by the truncation mechanism.

Step 2: Uniform control of the ordinary likelihood component.

Let

$$\mathcal{F}_n = \{\psi_{\beta, m} : \beta \in \mathcal{B}, m \in \mathcal{M}_n\}$$

denote the sieve class associated with the regression likelihood. By the entropy assumption imposed on the spline sieve and the bounded envelope assumption, there exists an integrable envelope F such that

$$|\psi_{\beta, m}(Y, X, Z)| \leq F(Y, X, Z), \quad PF^2 < \infty.$$

Moreover, the assumed entropy condition implies

$$\log N_{[]}(\epsilon, \mathcal{F}_n, L_2(P)) \leq AK_n \log(B/\epsilon)$$

for finite constants $A, B > 0$.

Since the sieve dimension satisfies the stated growth condition, the corresponding empirical-process bound gives

$$\sup_{\psi \in \mathcal{F}_n} |(P_n - P)\psi| = O_p \left(\sqrt{\frac{K_n \log n}{n}} \right) = o_p(1).$$

Hence,

$$\sup_{(\beta, m)} |(P_n - P)\psi_{\beta, m}| = o_p(1).$$

Step 3: Uniform control of the truncation-normalization component.

The second empirical-process term is specific to the proposed estimator. By the C -boundedness assumption, there exists a constant $c_0 > 0$ such that

$$C(X, Z; \beta, m) \geq c_0$$

uniformly over $(\beta, m) \in \mathcal{B} \times \mathcal{M}_n$. Together with the assumed upper bound on C , this gives

$$|\log C(X, Z; \beta, m)| \leq C_1$$

for some finite constant C_1 .

Define

$$\mathcal{C}_n = \{\rho_{\beta, m}(X, Z) = \log C(X, Z; \beta, m) : \beta \in \mathcal{B}, m \in \mathcal{M}_n\}.$$

The assumed smoothness of C implies that the class $\mathcal{C}_n$ inherits the entropy bound of the spline class. In particular,

$$\log N_{[]}(\epsilon, \mathcal{C}_n, L_2(P)) \leq A_1 K_n \log(B_1/\epsilon)$$

for finite constants $A_1, B_1 > 0$. Since $\mathcal{C}_n$ has a bounded square-integrable envelope, the empirical-process bound yields

$$\sup_{\rho \in \mathcal{C}_n} |(P_n - P)\rho| = O_p\left(\sqrt{\frac{K_n \log n}{n}}\right) = o_p(1).$$

Therefore,

$$\sup_{(\beta, m)} |(P_n - P) \log C(X, Z; \beta, m)| = o_p(1).$$

This is the additional empirical-process term that is absent from the ordinary untruncated partially nonlinear regression model.

Step 4: Control of the derivatives of the truncation normalization.

The truncation correction also enters the score through

$$\frac{\partial}{\partial \beta} \log C(X, Z; \beta, m) = \frac{C_\beta(X, Z; \beta, m)}{C(X, Z; \beta, m)}.$$

Since C is bounded away from zero and its derivatives are uniformly bounded by assumption,

$$\sup_{(\beta, m)} \left\| \frac{\partial}{\partial \beta} \log C(X, Z; \beta, m) \right\| \leq C_2$$

for some finite constant C_2 .

Similarly, for a perturbation

$$m_t = m + th,$$

the functional derivative is

$$\left. \frac{d}{dt} \log C(X, Z; \beta, m_t) \right|_{t=0} = \frac{C_m(X, Z; \beta, m)[h]}{C(X, Z; \beta, m)},$$

which is uniformly bounded under the assumed differentiability and C -boundedness conditions.

The corresponding derivative classes satisfy the same entropy and envelope conditions. Hence, their empirical processes are stochastically equicontinuous. In particular,

$$\sup_{(\beta, m)} \left| (P_n - P) \frac{\partial}{\partial \beta} \log C(X, Z; \beta, m) \right| = o_p(1),$$

and

$$\sup_{(\beta, m)} \left| (P_n - P) \left[\frac{C_m[h]}{C} \right] \right| = o_p(1).$$

Thus, the parameter-dependent normalization does not merely add a bounded deterministic correction; it generates additional stochastic terms in both the parametric and nonparametric estimating equations. The stated entropy and stochastic-equicontinuity assumptions ensure that these terms remain uniformly negligible.

Step 5: Uniform convergence of the criterion.

Combining Steps 2 and 3 gives

$$\sup_{(\beta, m)} |M_n(\beta, m) - M(\beta, m)| = o_p(1).$$

Thus,

$$\|M_n - M\|_\infty \xrightarrow{p} 0.$$

Step 6: Spline approximation.

Let $\mathcal{M}_n$ denote the spline sieve space and let $m_{0,n} \in \mathcal{M}_n$ be the spline approximation to m_0 . Under the assumed smoothness condition on m_0 ,

$$\|m_{0,n} - m_0\|_{L_2(P_Z)} = O(K_n^{-r}).$$

Since $K_n \rightarrow \infty$,

$$\|m_{0,n} - m_0\|_{L_2(P_Z)} \rightarrow 0.$$

The truncation normalization preserves this approximation property. Indeed, by the boundedness of the derivative of $\log C$ with respect to m ,

$$|\log C(X, Z; \beta, m_{0,n}) - \log C(X, Z; \beta, m_0)| \leq C_3 \|m_{0,n} - m_0\|_{L_2(P_Z)}$$

for some finite constant C_3 . Consequently,

$$\sup_{\beta \in \mathcal{B}} |P \log C(X, Z; \beta, m_{0,n}) - P \log C(X, Z; \beta, m_0)| = o(1).$$

Step 7: Identification of the population maximizer.

By the identifiability assumption, the population criterion has a unique maximizer at (β_0, m_0) . Hence, for every $\epsilon > 0$, there exists $\eta_\epsilon > 0$ such that

$$\sup_{\|\beta - \beta_0\| + \|m - m_0\|_{L_2(P_Z)} \geq \epsilon} M(\beta, m) \leq M(\beta_0, m_0) - \eta_\epsilon.$$

Because

$$\sup_{(\beta, m)} |M_n(\beta, m) - M(\beta, m)| = o_p(1),$$

and because the spline approximation error tends to zero, with probability tending to one,

$$M_n(\hat{\beta}, \hat{m}) > M_n(\beta_0, m_{0,n}) - \frac{\eta_\epsilon}{2}.$$

Therefore, the maximizer cannot remain outside an ϵ -neighborhood of (β_0, m_0) . It follows that

$$\|\hat{\beta} - \beta_0\| + \|\hat{m} - m_0\|_{L_2(P_Z)} \xrightarrow{p} 0.$$

Hence,

$$\hat{\beta} \xrightarrow{p} \beta_0, \quad \|\hat{m} - m_0\|_{L_2(P_Z)} \xrightarrow{p} 0.$$

The proof explicitly identifies the additional terms introduced by truncation, namely the empirical process generated by $\log C(X, Z; \beta, m)$ and the corresponding empirical processes generated by its parametric and functional derivatives. The lower bound on C , together with the entropy, envelope, differentiability, and stochastic-equicontinuity assumptions, provides uniform control of these terms. Therefore, the consistency argument extends the standard sieve M-estimation framework while explicitly accounting for the parameter-dependent truncation normalization.

This completes the proof. $\square$

A.5 Proof of Theorem 3.5

Proof. We decompose the estimation error into the spline approximation error and the stochastic estimation error:

$$\hat{m} - m_0 = (\hat{m} - m_{K_n}) + (m_{K_n} - m_0),$$

where m_{K_n} denotes the best spline approximation to m_0 in the K_n -dimensional spline sieve space $\mathcal{M}_n$.

Step 1: Spline approximation error.

Since m_0 belongs to an r -smooth function class, the standard spline approximation property gives

$$\|m_{K_n} - m_0\|_\infty = O(K_n^{-r}),$$

and therefore

$$\|m_{K_n} - m_0\|_{L_2(P_Z)} = O(K_n^{-r}).$$

Thus, it remains to establish the stochastic order of $\|\hat{m} - m_{K_n}\|_{L_2(P_Z)}$.

Step 2: Spline representation and score expansion.

Let

$$m_\theta(z) = B_{K_n}(z)^\top \theta,$$

where $B_{K_n}(z)$ is the K_n -dimensional spline basis, and let θ_{K_n} denote the coefficient vector corresponding to m_{K_n} . Define the penalized criterion

$$Q_n(\beta, \theta) = \ell_n(\beta, \theta) - \lambda_n \theta^\top \Omega_n \theta.$$

The score with respect to θ is

$$S_{\theta,n}(\beta, \theta) = \frac{\partial \ell_n(\beta, \theta)}{\partial \theta} - 2\lambda_n \Omega_n \theta.$$

The likelihood contribution for observation i is

$$\ell_i(\beta, \theta) = \log f_\varepsilon \{Y_i - m_\theta(Z_i) - g(X_i; \beta)\} - \log C(X_i, Z_i; \beta, m_\theta).$$

Consequently, the spline score contains the ordinary likelihood component together with the truncation-specific term

$$- \frac{D_m C(X_i, Z_i; \beta, m_\theta)[B_{K_n}(Z_i)]}{C(X_i, Z_i; \beta, m_\theta)}.$$

Thus,

$$S_{\theta,n} = S_{\theta,n}^{(0)} + S_{\theta,n}^{(C)} - 2\lambda_n \Omega_n \theta,$$

where $S_{\theta,n}^{(0)}$ denotes the ordinary regression score and $S_{\theta,n}^{(C)}$ is generated by the truncation normalization.

By the assumption

$$\inf_{x,z,\beta,m} C(x, z; \beta, m) \geq c_C > 0,$$

and the assumed boundedness of the derivatives of C , the additional term $S_{\theta,n}^{(C)}$ is uniformly controlled. In particular, the entropy and envelope conditions imposed on the corresponding score class imply

$$\|S_{\theta,n}^{(C)} - ES_{\theta,n}^{(C)}\| = O_p(\sqrt{nK_n}).$$

The same order holds for the ordinary score component. Hence, at (β_0, θ_{K_n}) ,

$$\|S_{\theta,n}(\beta_0, \theta_{K_n})\| = O_p(\sqrt{nK_n}) + O(n\lambda_n).$$

Under the assumed conditions on λ_n , the deterministic penalty component is of smaller order for the function-level rate considered below. Therefore, the dominant stochastic score satisfies

$$\|S_{\theta,n}(\beta_0, \theta_{K_n})\| = O_p(\sqrt{nK_n}).$$

Step 3: Taylor expansion of the penalized score.

Since

$$S_{\theta,n}(\hat{\beta}, \hat{\theta}) = 0,$$

a Taylor expansion around (β_0, θ_{K_n}) gives

$$0 = S_{\theta,n}(\beta_0, \theta_{K_n}) - H_{\theta\theta,n}(\tilde{\beta}, \tilde{\theta})(\hat{\theta} - \theta_{K_n}) \\ - H_{\theta\beta,n}(\tilde{\beta}, \tilde{\theta})(\hat{\beta} - \beta_0) - 2\lambda_n\Omega_n\hat{\theta} + R_n,$$

where $(\tilde{\beta}, \tilde{\theta})$ lies between (β_0, θ_{K_n}) and $(\hat{\beta}, \hat{\theta})$, and R_n collects the higher-order Taylor terms.

The Hessian has two components,

$$H_{\theta\theta,n} = H_{\theta\theta,n}^{(0)} + H_{\theta\theta,n}^{(C)} + 2\lambda_n\Omega_n,$$

where $H_{\theta\theta,n}^{(C)}$ contains the second derivatives of $\log C$. In particular, these terms involve

$$\frac{D_{mm}C}{C} - \frac{D_m C D_m C^\top}{C^2}.$$

The lower bound on C and the assumed boundedness of C , $D_m C$, and $D_{mm}C$ ensure that these truncation-specific Hessian terms are uniformly bounded.

By the entropy and stochastic-equicontinuity assumptions,

$$\sup_{\|(\beta,\theta) - (\beta_0, \theta_{K_n})\| \leq \delta_n} \left\| \frac{1}{n} H_{\theta\theta,n} - \mathcal{I}_{\theta\theta} \right\| = o_p(1),$$

for any sequence $\delta_n \rightarrow 0$. Hence,

$$H_{\theta\theta,n} = n\mathcal{I}_{\theta\theta} + o_p(n),$$

where $\mathcal{I}_{\theta\theta}$ is the truncation-adjusted population information matrix restricted to the spline sieve.

By the positive-definiteness assumption,

$$\lambda_{\min}(\mathcal{I}_{\theta\theta}) \geq c_I > 0.$$

Therefore,

$$\lambda_{\min}(H_{\theta\theta,n}) \geq \frac{c_I}{2}n$$

with probability tending to one, and hence

$$\|H_{\theta\theta,n}^{-1}\| = O_p(n^{-1}).$$

Step 4: Control of the nuisance and remainder terms.

The cross-Hessian term satisfies

$$\|H_{\theta\beta,n}\| = O_p(n),$$

while, by Theorem 3.4,

$$\|\hat{\beta} - \beta_0\| = o_p(1).$$

The stochastic equicontinuity assumption ensures that the resulting local contribution is of smaller order than the leading spline score after multiplication by $H_{\theta\theta,n}^{-1}$. Similarly, the Taylor remainder satisfies

$$\|R_n\| = o_p\left(\sqrt{nK_n}\right) + o_p\left(n\|\hat{\theta} - \theta_{K_n}\|\right).$$

The penalty contribution satisfies

$$\left\|2\lambda_n\Omega_n\hat{\theta}\right\| = O_p(n\lambda_n)$$

under the stated bounds on the spline coefficients and penalty matrix. Under the assumed tuning conditions, its contribution to the function-level estimation error is asymptotically negligible relative to the stochastic spline estimation term.

Thus, after premultiplication by $H_{\theta\theta,n}^{-1}$,

$$\hat{\theta} - \theta_{K_n} = H_{\theta\theta,n}^{-1}S_{\theta,n}(\beta_0, \theta_{K_n}) + o_p\left(\sqrt{\frac{K_n}{n}}\right).$$

Since

$$\|H_{\theta\theta,n}^{-1}\| = O_p(n^{-1})$$

and

$$\|S_{\theta,n}(\beta_0, \theta_{K_n})\| = O_p(\sqrt{nK_n}),$$

we obtain

$$\|\hat{\theta} - \theta_{K_n}\| = O_p\left(\sqrt{\frac{K_n}{n}}\right).$$

Step 5: Conversion from coefficient error to function error.

Because

$$\hat{m}(z) - m_{K_n}(z) = B_{K_n}(z)^\top(\hat{\theta} - \theta_{K_n}),$$

the assumed boundedness and regularity of the spline basis imply

$$\|\hat{m} - m_{K_n}\|_{L_2(P_Z)} = O_p\left(\sqrt{\frac{K_n}{n}}\right).$$

Combining this stochastic error with the approximation error gives

$$\begin{aligned}\|\hat{m} - m_0\|_{L_2(P_Z)} &\leq \|\hat{m} - m_{K_n}\|_{L_2(P_Z)} + \|m_{K_n} - m_0\|_{L_2(P_Z)} \\ &= O_p\left(\sqrt{\frac{K_n}{n}}\right) + O(K_n^{-r}).\end{aligned}$$

Therefore,

$$\boxed{\|\hat{m} - m_0\|_{L_2(P_Z)} = O_p\left(K_n^{-r} + \sqrt{\frac{K_n}{n}}\right)}.$$

The proof explicitly shows the role of the truncation mechanism in the rate derivation. In addition to the ordinary spline score and Hessian, the derivatives

$$D_m \log C = \frac{D_m C}{C}$$

and

$$D_{mm} \log C = \frac{D_{mm} C}{C} - \frac{D_m C D_m C^\top}{C^2}$$

enter the stochastic score and information matrix. The uniform lower bound on C prevents these terms from becoming unbounded, while the entropy and stochastic-equicontinuity assumptions control their empirical-process fluctuations uniformly over the growing spline sieve. Consequently, the truncation correction modifies the constants and the information operator but does not introduce an additional dominant term in the convergence rate.

Hence, under Assumptions 1–5, the proposed estimator attains the rate

$$K_n^{-r} + \sqrt{K_n/n}.$$

□

A.6 Proof of Theorem 3.6

Proof. Let

$$S_{\beta,n}(\beta, m) = \frac{\partial \ell_n(\beta, m)}{\partial \beta}$$

denote the score function with respect to the finite-dimensional parameter β . From (2.4),

$$S_{\beta,n}(\beta, m) = \sum_{i=1}^n \psi_{\beta,i}(\beta, m),$$

where

$$\psi_{\beta,i}(\beta, m) = \frac{\partial}{\partial \beta} \log f_\varepsilon \{Y_i - m(Z_i) - g(X_i; \beta)\} - \frac{\partial}{\partial \beta} \log C(X_i, Z_i; \beta, m).$$

The second term is specific to the truncation mechanism and is absent under ordinary untruncated sampling.

Let

$$\psi_{\beta,i}^0 = \psi_{\beta,i}(\beta_0, m_0).$$

Since the likelihood is correctly specified under the conditional truncated sampling scheme,

$$E\{\psi_{\beta,i}^0\} = 0.$$

By the truncation-stability assumption, there exists $c_C > 0$ such that

$$C(X, Z; \beta, m) \geq c_C$$

uniformly over the relevant parameter and sieve spaces. Hence, if the corresponding derivatives of C are bounded, then

$$\sup_{\beta, m} \left\| \frac{\partial}{\partial \beta} \log C(X, Z; \beta, m) \right\| < \infty,$$

and

$$\sup_{\beta, m} \left\| \frac{\partial^2}{\partial \beta \partial \beta^\top} \log C(X, Z; \beta, m) \right\| < \infty.$$

These bounds are important because they ensure that the additional truncation-dependent score and information terms do not dominate the usual likelihood terms.

We first control the effect of estimating the nuisance function. By Theorem 3.5,

$$\|\hat{m} - m_0\| = O_p(n^{-r/(2r+1)}).$$

For $r \geq 2$,

$$n^{-r/(2r+1)} = o(n^{-1/4}),$$

and therefore

$$\|\hat{m} - m_0\| = o_p(n^{-1/4}).$$

We next expand the score around (β_0, m_0) . Since $S_{\beta, n}(\hat{\beta}, \hat{m}) = 0$, a Taylor expansion gives

$$\begin{aligned} 0 &= S_{\beta, n}(\beta_0, m_0) + \frac{\partial S_{\beta, n}}{\partial \beta} \Big|_{(\beta_0, m_0)} (\hat{\beta} - \beta_0) \\ &\quad + \frac{\partial S_{\beta, n}}{\partial m} \Big|_{(\beta_0, m_0)} [\hat{m} - m_0] + R_{\beta, n}, \end{aligned}$$

where $R_{\beta, n}$ contains the second- and higher-order remainder terms.

Define

$$I_n(\beta, m) = -\frac{\partial S_{\beta, n}(\beta, m)}{\partial \beta}.$$

Then

$$\begin{aligned} I_n(\beta, m) &= -\sum_{i=1}^n \frac{\partial^2}{\partial \beta \partial \beta^\top} \log f_\varepsilon \{Y_i - m(Z_i) - g(X_i; \beta)\} \\ &\quad + \sum_{i=1}^n \frac{\partial^2}{\partial \beta \partial \beta^\top} \log C(X_i, Z_i; \beta, m). \end{aligned}$$

Thus, the observed information contains an additional truncation-dependent component involving the second derivative of $\log C$.

The derivative of the score with respect to the nuisance function also contains a truncation-specific term. For a perturbation $m_t = m + th$, we have

$$\begin{aligned} \frac{\partial S_{\beta, n}}{\partial m}[h] &= -\sum_{i=1}^n \frac{\partial^2}{\partial \beta \partial m} \log f_\varepsilon \{Y_i - m(Z_i) - g(X_i; \beta)\} [h] \\ &\quad - \sum_{i=1}^n \frac{\partial}{\partial \beta} \left\{ \frac{d}{dt} \log C(X_i, Z_i; \beta, m_t) \Big|_{t=0} \right\}. \end{aligned}$$

The second term is absent in the conventional untruncated model and represents the additional empirical-process component induced by joint truncation.

Under the entropy and envelope assumptions imposed on the sieve class, the classes generated by

$$\frac{\partial}{\partial \beta} \log C(X, Z; \beta, m), \quad \frac{\partial^2}{\partial \beta \partial \beta^\top} \log C(X, Z; \beta, m),$$

and

$$\frac{\partial}{\partial \beta} \left\{ \frac{d}{dt} \log C(X, Z; \beta, m_t) \right\}_{t=0}$$

possess integrable envelopes and satisfy the required uniform entropy bounds. Consequently, the corresponding empirical processes are stochastically equicontinuous. In particular,

$$\sup_{m \in \mathcal{M}_n} \left| (P_n - P) \frac{\partial}{\partial \beta} \log C(X, Z; \beta_0, m) \right| = o_p(n^{-1/2}),$$

and analogous bounds hold for the second-order derivative classes.

Therefore, the nuisance-function contribution satisfies

$$\frac{1}{\sqrt{n}} \frac{\partial S_{\beta,n}}{\partial m} [\widehat{m} - m_0] = o_p(1),$$

after using the $o_p(n^{-1/4})$ nuisance rate together with the orthogonality of the score and the stochastic equicontinuity of the relevant empirical-process classes. The second-order Taylor remainder satisfies

$$\frac{1}{\sqrt{n}} R_{\beta,n} = o_p(1).$$

Hence,

$$0 = \frac{1}{\sqrt{n}} S_{\beta,n}(\beta_0, m_0) - \frac{I_n(\beta_0, m_0)}{n} \sqrt{n}(\widehat{\beta} - \beta_0) + o_p(1).$$

We now establish convergence of the observed information. By the uniform law of large numbers applied jointly to the ordinary likelihood derivative class and the truncation-dependent class,

$$\frac{1}{n} I_n(\beta_0, m_0) \xrightarrow{p} I(\beta_0),$$

where

$$\begin{aligned} I(\beta_0) = & -E \left[\frac{\partial^2}{\partial \beta \partial \beta^\top} \log f_\varepsilon \{Y - m_0(Z) - g(X; \beta_0)\} \right] \\ & + E \left[\frac{\partial^2}{\partial \beta \partial \beta^\top} \log C(X, Z; \beta_0, m_0) \right]. \end{aligned}$$

The second expectation is the additional information contribution resulting from the parameter-dependent truncation normalization. By assumption, $I(\beta_0)$ is positive definite.

It remains to establish the limiting distribution of the score. At the true parameter values,

$$S_{\beta,n}(\beta_0, m_0) = \sum_{i=1}^n \psi_{\beta,i}^0,$$

where

$$\psi_{\beta,i}^0 = \frac{\partial}{\partial \beta} \log f_\varepsilon \{Y_i - m_0(Z_i) - g(X_i; \beta_0)\} - \frac{\partial}{\partial \beta} \log C(X_i, Z_i; \beta_0, m_0).$$

The observed sample is i.i.d. under the conditional truncated sampling scheme. The lower bound on C and the assumed boundedness of its derivatives imply that the truncation-dependent component of $\psi_{\beta,i}^0$ has finite second moment. Together with the assumed moment condition for the ordinary likelihood score,

$$E\|\psi_{\beta,i}^0\|^2 < \infty.$$

Hence, the Lindeberg condition is satisfied. The Lindeberg–Feller central limit theorem therefore gives

$$\frac{1}{\sqrt{n}} S_{\beta,n}(\beta_0, m_0) \xrightarrow{d} N\{0, \Sigma(\beta_0)\},$$

where

$$\Sigma(\beta_0) = \text{Var}\{\psi_{\beta,i}^0\}.$$

Under the correctly specified likelihood, the information identity gives

$$\Sigma(\beta_0) = I(\beta_0).$$

More generally, without imposing the information identity, the asymptotic covariance matrix is the sandwich matrix

$$I(\beta_0)^{-1} \Sigma(\beta_0) I(\beta_0)^{-1}.$$

Combining the score expansion, convergence of the observed information, and the central limit theorem yields

$$\sqrt{n}(\hat{\beta} - \beta_0) = I(\beta_0)^{-1} \frac{1}{\sqrt{n}} S_{\beta,n}(\beta_0, m_0) + o_p(1).$$

Consequently,

$$\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} N\{0, I(\beta_0)^{-1} \Sigma(\beta_0) I(\beta_0)^{-1}\}.$$

Under the information identity,

$$\boxed{\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} N\{0, I(\beta_0)^{-1}\}}.$$

The proof identifies the additional technical difficulty created by truncation. Apart from the standard sieve-M-estimation terms, the empirical-process analysis must control the score, Hessian, and nuisance derivatives generated by the parameter-dependent quantity $\log C(X, Z; \beta, m)$. The uniform lower bound on C , together with the entropy, envelope, smoothness, and stochastic-equicontinuity conditions, ensures that these additional terms are asymptotically negligible where required and converge uniformly in the information expansion. Thus, the asymptotic normality result follows only after explicitly accounting for the truncation-dependent normalization.

□

A.7 Proof of Theorem 3.7

Proof. The result follows by examining the three components that determine the stability and consistency of the sieve estimator: the approximation error, the stochastic estimation error, and the truncation-adjusted likelihood contribution. Recall from Theorem 3.6 that the estimation error of the nonlinear component admits the decomposition

$$\|\hat{m} - m_0\| = O_p \left(K_n^{-r} + \sqrt{\frac{K_n}{n}} + \mathcal{R}_{n,\lambda} \right),$$

where K_n^{-r} is the spline approximation error, $\sqrt{K_n/n}$ is the stochastic estimation error associated with the increasing sieve dimension, and $\mathcal{R}_{n,\lambda}$ denotes the additional contribution induced by penalization and the truncation-adjusted likelihood. We show that each of the conditions stated in the theorem is necessary for these terms to remain asymptotically negligible.

Case 1: $K_n/n \not\rightarrow 0$.

Suppose first that

$$\frac{K_n}{n} \not\rightarrow 0.$$

Then there exists a subsequence, denoted again by n , and a constant $c > 0$ such that

$$\frac{K_n}{n} \geq c$$

along that subsequence. Consequently,

$$\sqrt{\frac{K_n}{n}} \geq \sqrt{c} > 0.$$

Hence the stochastic component of the sieve estimation error does not converge to zero. More precisely, the empirical-process contribution to the criterion satisfies

$$\sup_{\theta \in \Theta_{K_n}} |(\mathbb{P}_n - P) \{\ell_\theta - \ell_{\theta_0}\}| = O_p \left(\sqrt{\frac{K_n}{n}} \right),$$

where $\mathbb{P}_n$ denotes the empirical measure and P the corresponding population expectation. Since the right-hand side is bounded away from zero, the stochastic perturbation of the population criterion does not disappear.

On the other hand, the spline approximation component satisfies

$$\inf_{m_{K_n} \in \mathcal{M}_{K_n}} \|m_{K_n} - m_0\| = O(K_n^{-r}),$$

which tends to zero whenever $K_n \rightarrow \infty$. Thus increasing the sieve dimension can reduce the approximation bias, but if K_n grows at a rate comparable to or faster than n , the reduction in approximation error is offset by the increasing stochastic variability.

Indeed, from the basic error decomposition,

$$\|\hat{m} - m_0\| \geq \text{stochastic contribution} - \text{remaining terms},$$

and the stochastic contribution cannot vanish when $K_n/n \not\rightarrow 0$. Therefore,

$$\|\hat{m} - m_0\| \not\rightarrow_p 0.$$

Thus the condition

$$\frac{K_n}{n} \rightarrow 0$$

is required to ensure that the effective dimension of the sieve does not increase too rapidly relative to the available sample size.

Case 2: $n\lambda_n \not\rightarrow \infty$.

Consider next the penalty parameter λ_n . The penalized objective function can be written schematically as

$$Q_n(\theta) = -\ell_n(\theta) + \lambda_n \theta^\top \Omega_n \theta,$$

where Ω_n is the spline penalty matrix. Its Hessian therefore has the form

$$\nabla^2 Q_n(\theta) = -\nabla^2 \ell_n(\theta) + 2\lambda_n \Omega_n.$$

The first component represents the information supplied by the observed data, whereas the second component provides regularization of the spline coefficients.

Suppose that

$$n\lambda_n \not\rightarrow \infty.$$

Then there exists a subsequence for which $n\lambda_n$ is bounded. Hence the regularization contribution

$$2\lambda_n \Omega_n$$

does not increase sufficiently fast relative to the stochastic variation associated with the increasing-dimensional parameter vector.

Let θ_{K_n} denote the best spline approximation to the true parameter. A local expansion of the penalized criterion around θ_{K_n} gives

$$0 = \nabla Q_n(\hat{\theta}) = \nabla Q_n(\theta_{K_n}) + \nabla^2 Q_n(\tilde{\theta})(\hat{\theta} - \theta_{K_n}),$$

for some intermediate point $\tilde{\theta}$. Therefore,

$$\hat{\theta} - \theta_{K_n} = -\left\{\nabla^2 Q_n(\tilde{\theta})\right\}^{-1} \nabla Q_n(\theta_{K_n}).$$

The stochastic score term is of order

$$O_p\left(\sqrt{\frac{K_n}{n}}\right),$$

while the penalty contributes the deterministic term

$$2\lambda_n \Omega_n \theta_{K_n}.$$

If $n\lambda_n$ does not diverge, the penalty is insufficient to provide the uniform curvature required to control the increasingly large spline parameter vector. Consequently, the minimum eigenvalue of the penalized Hessian cannot be guaranteed to remain bounded away from zero:

$$\inf_{\theta \in \Theta_{K_n}} \lambda_{\min} \{ \nabla^2 Q_n(\theta) \} \not\geq c_0 > 0.$$

The inverse Hessian can therefore become arbitrarily large, and the stochastic error

$$\|\hat{\theta} - \theta_{K_n}\| \leq \left\| \nabla^2 Q_n(\tilde{\theta})^{-1} \right\| \|\nabla Q_n(\theta_{K_n})\|$$

cannot be uniformly controlled.

As a consequence, the spline coefficient estimator may exhibit increasing variance and instability. Hence the estimation error of the nonlinear component does not necessarily converge to zero:

$$\|\hat{m} - m_0\| \not\rightarrow_p 0.$$

Therefore, the condition

$$n\lambda_n \rightarrow \infty$$

is required to maintain sufficient regularization and uniform control of the penalized criterion.

Case 3: $\inf_{x,z} C(x, z; \beta, m) = 0$.

Finally, consider the truncation-normalization term. The truncated conditional density is

$$f^{(T)}(y \mid x, z; \beta, m) = \frac{f_\varepsilon\{y - m(z) - g(x; \beta)\}}{C(x, z; \beta, m)},$$

where

$$C(x, z; \beta, m) = P(a_Y \leq Y \leq b_Y \mid X = x, Z = z).$$

Consequently, the log-likelihood contains the additional term

$$-\log C(x, z; \beta, m).$$

The corresponding parametric score therefore contains

$$-\frac{\partial}{\partial \beta} \log C(x, z; \beta, m) = -\frac{\partial C(x, z; \beta, m) / \partial \beta}{C(x, z; \beta, m)}.$$

Similarly, the derivative with respect to the nonlinear component contains terms of the form

$$-\frac{D_m C(x, z; \beta, m)[h]}{C(x, z; \beta, m)},$$

for perturbations h of m . Thus, unlike the ordinary untruncated likelihood, the score contains inverse powers of the normalization constant C .

Suppose now that

$$\inf_{x,z} C(x, z; \beta, m) = 0.$$

Then, for every $M > 0$, there exist values (x, z) such that

$$\frac{1}{C(x, z; \beta, m)} > M.$$

Therefore, the truncation-adjusted score contributions

$$\frac{\partial}{\partial \beta} \log C(X_i, Z_i; \beta, m)$$

can become arbitrarily large. The same phenomenon occurs for the functional derivative involving $D_m C$.

In particular, if the derivatives of C remain nonzero in regions where C approaches zero, then

$$\left| \frac{D_\beta C}{C} \right| \rightarrow \infty, \quad \left| \frac{D_m C[h]}{C} \right| \rightarrow \infty.$$

Consequently, the envelope required for uniform control of the empirical process,

$$\sup_{\theta \in \Theta_{K_n}} |(\mathbb{P}_n - P)\psi_\theta|,$$

where ψ_θ denotes the truncation-adjusted score contribution, need not be finite.

The same problem appears in the second derivative. Differentiating $\log C$ twice gives terms of the form

$$\frac{D^2 C}{C} - \frac{(DC)(DC)^\top}{C^2}.$$

Hence, as $C \rightarrow 0$, the Hessian contributions may diverge at rates of order C^{-1} or C^{-2} . The information matrix can consequently fail to be uniformly bounded and nonsingular. In particular, the regularity conditions required for the uniform law of large numbers and for the central limit theorem for the score are no longer guaranteed.

This demonstrates why the lower-bound condition

$$\inf_{x, z} C(x, z; \beta, m) \geq c_C > 0$$

is essential. It provides the uniform bound

$$\frac{1}{C(x, z; \beta, m)} \leq \frac{1}{c_C},$$

which in turn controls the truncation-adjusted score and its derivatives. It also prevents the likelihood normalization from amplifying stochastic fluctuations in regions where the probability of being observed is very small.

Conclusion: The three conditions control distinct components of the estimator. The condition $K_n/n \rightarrow 0$ controls the stochastic complexity of the growing sieve, the condition $n\lambda_n \rightarrow \infty$ guarantees sufficient regularization and curvature of the penalized criterion, and the condition

$$\inf_{x, z} C(x, z; \beta, m) > 0$$

controls the additional stochastic terms generated by the truncation-dependent normalization.

When these conditions hold, the approximation error K_n^{-r} , the stochastic error $\sqrt{K_n/n}$, and the penalization/truncation contributions can be controlled uniformly, which allows the consistency result established in Theorem 3.6 to hold. If any one of the conditions fails, at least one of the required components is no longer asymptotically negligible or uniformly controllable. Consequently, the convergence

$$\|\hat{m} - m_0\| \rightarrow_p 0$$

cannot, in general, be guaranteed.

□